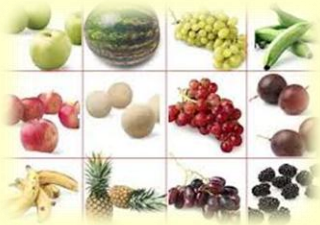


Validation Model of Classification



Ali Ridho Barakbah

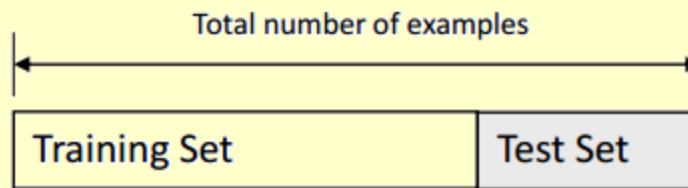
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Validation Model of Classification

- Holdout method
- K-fold cross validation
- Leave-one-out cross validation

Holdout Method

- Split dataset into two groups
 - Training set: used to train the classifier
 - Test set: used to estimate the error rate of the trained classifier

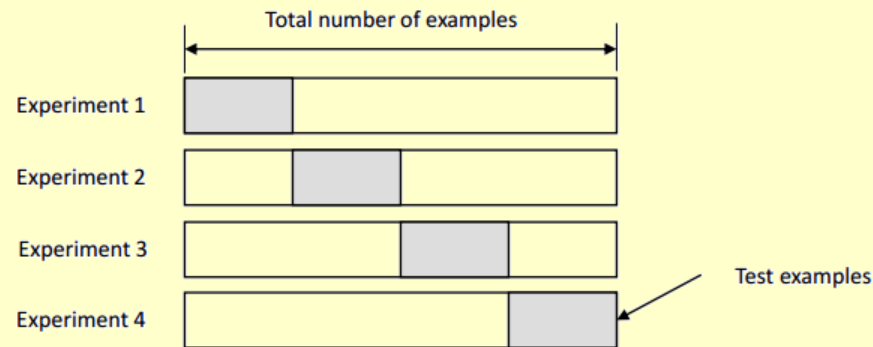


- The holdout method has two basic drawbacks
 - In problems where we have a sparse dataset we may not be able to afford the “luxury” of setting aside a portion of the dataset for testing
 - Since it is a single train-and-test experiment, the holdout estimate of error rate will be misleading if we happen to get an “unfortunate” split

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K-fold Cross Validation

- Create a K-fold partition of the dataset
 - For each of K experiments, use $K - 1$ folds for training and a different fold for testing
 - This procedure is illustrated in the following figure for $K = 4$



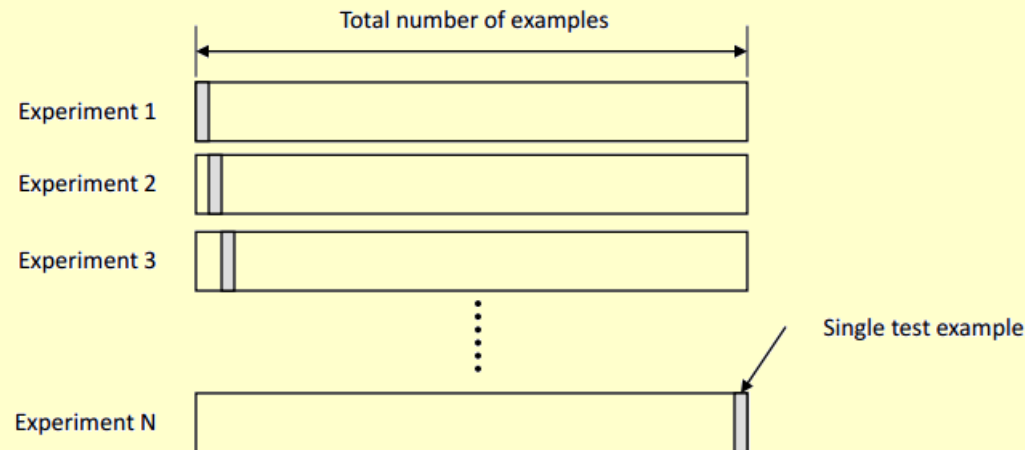
- K-Fold cross validation is similar to random subsampling
 - The advantage of KFCV is that all the examples in the dataset are eventually used for both training and testing
 - As before, the true error is estimated as the average error rate on test examples

$$E = \frac{1}{K} \sum_{i=1}^K E_i$$

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Leave-one-out Cross Validation

- LOO is the degenerate case of KFCV, where K is chosen as the total number of examples
 - For a dataset with N examples, perform ?? Experiments
 - For each experiment use $N - 1$ examples for training and the remaining example for testing



- As usual, the true error is estimated as the average error rate on test examples

$$E = \frac{1}{N} \sum_{i=1}^N E_i$$

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How many folds are needed?

- With a large number of folds
 - + The bias of the true error rate estimator will be small (the estimator will be very accurate)
 - The variance of the true error rate estimator will be large
 - The computational time will be very large as well (many experiments)
- With a small number of folds
 - + The number of experiments and, therefore, computation time are reduced
 - + The variance of the estimator will be small
 - The bias of the estimator will be large (conservative or larger than the true error rate)
- In practice, the choice for K depends on the size of the dataset
 - For large datasets, even 3-fold cross validation will be quite accurate
 - For very sparse datasets, we may have to use leave-one-out in order to train on as many examples as possible
- A common choice for is $K=10$

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